

Rapport – Mini-projet : Supervision avec Wazuh

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Module : Cybersécurité

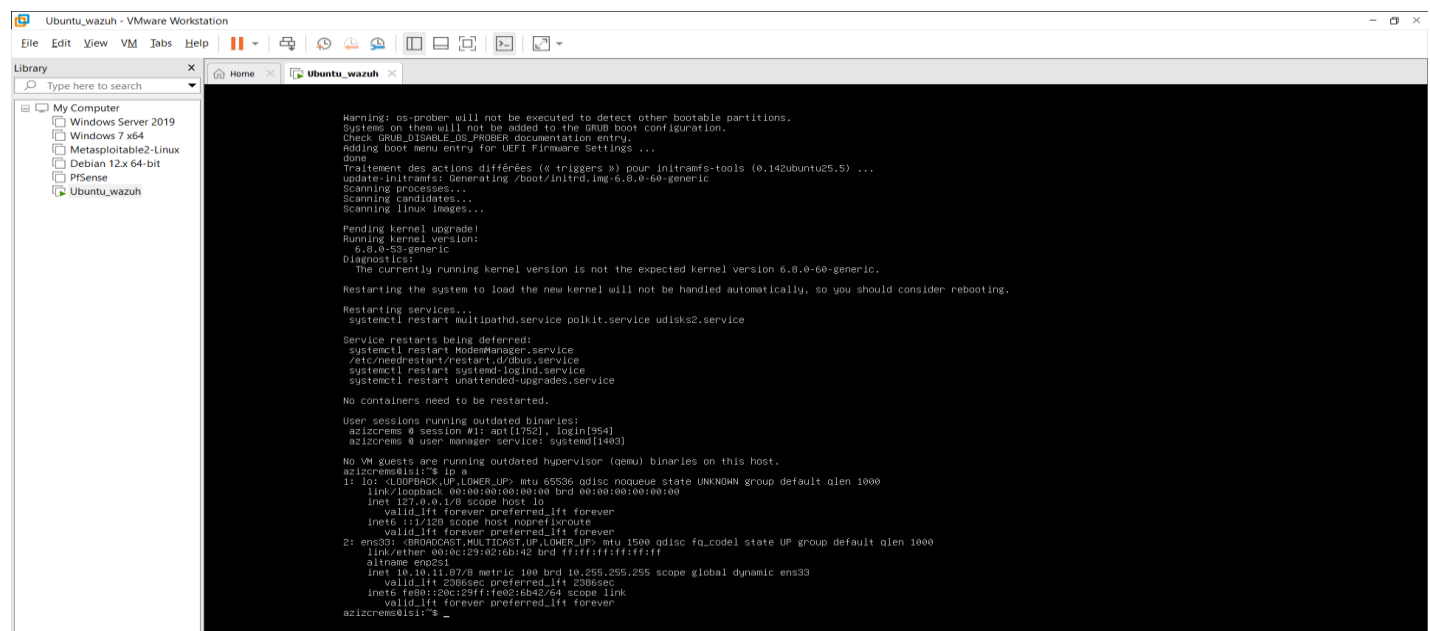
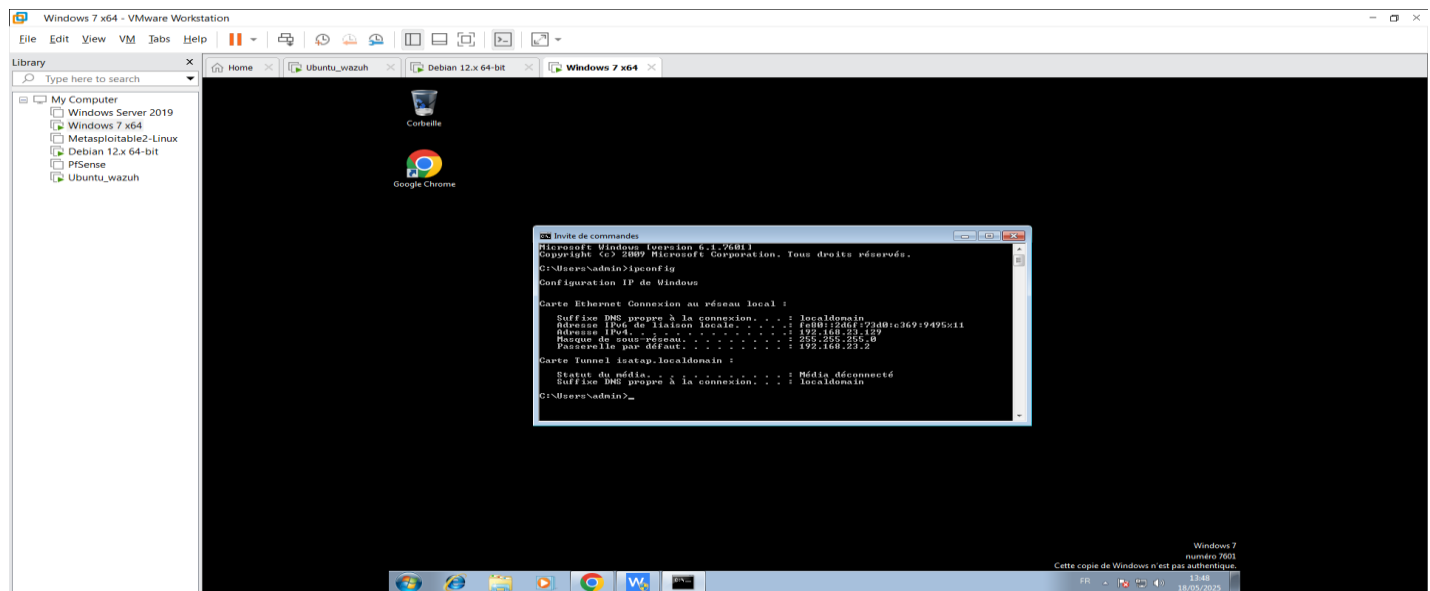
Encadrant : Mme Chaimae KISSI

1. Objectif du TP

- Mettre en place un serveur de supervision Wazuh sous Ubuntu.
- Déployer un agent sur une machine Windows.
- Générer et analyser des alertes de sécurité dans l'interface Dashboard Wazuh.

2. Topologie du réseau

- 1 VM Ubuntu 22.04 (Wazuh Manager)
- 1 VM Windows 7 x64 (Agent supervisé)
- IP Wazuh Manager : 192.168.23.135
- IP Agent Windows : 192.168.23.192
- Réseau NAT ou Host-only sur VMware



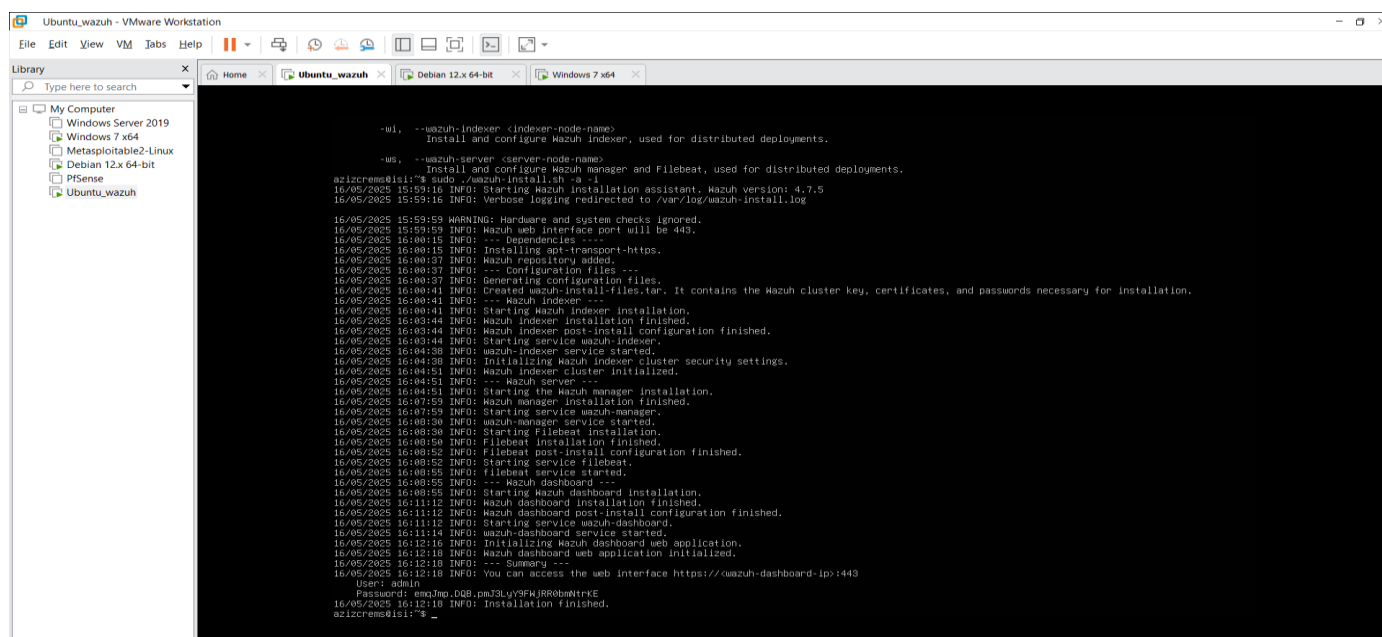
3. Installation du serveur Wazuh (Ubuntu)

Commandes exécutées :

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
chmod +x ./wazuh-install.sh
sudo ./wazuh-install.sh -a --ignore-os-check
```

Problèmes rencontrés : système non recommandé

→ solution : ajout de l'option '--ignore-os-check'



```
Ubuntu_wazuh - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Windows Server 2019
Windows 7 x64
Metasploit2-Linux
Debian 12.x 64-bit
PISense
Ubuntu_wazuh

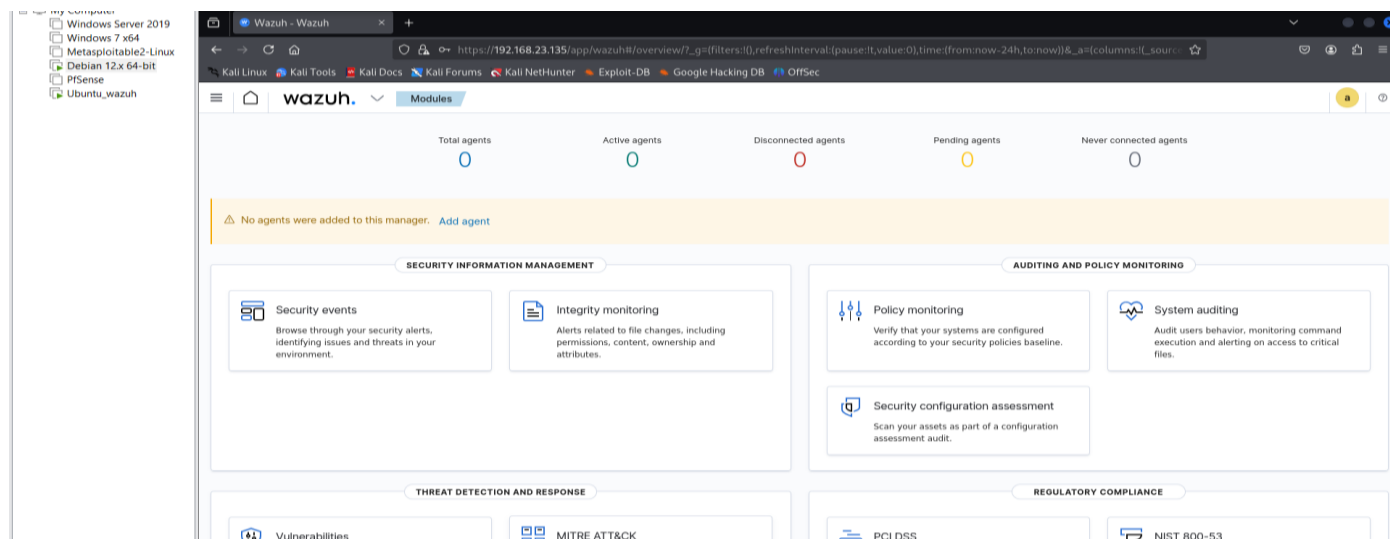
-wi, --wazuh-indexer <indexer-node-name>
Install and configure Wazuh indexer, used for distributed deployments.

-ws, --wazuh-server <server-node-name>
Install and configure Wazuh manager and Filebeat, used for distributed deployments.

az12cems@isi:~$ sudo ./wazuh-install.sh -a -i
16/05/2025 15:53:16 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
16/05/2025 15:53:16 INFO: Verbose logging redirected to /var/log/wazuh-install.log
16/05/2025 15:53:59 WARNING: Hardware and system checks ignored.
16/05/2025 15:53:59 INFO: Wazuh web interface port will be 443.
16/05/2025 16:00:15 INFO: -- Dependencies --
16/05/2025 16:00:15 INFO: Installing apt-transport-https.
16/05/2025 16:00:37 INFO: Wazuh repository added.
16/05/2025 16:00:37 INFO: -- Configuration files --
16/05/2025 16:00:37 INFO: Generating configuration files.
16/05/2025 16:00:41 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
16/05/2025 16:00:41 INFO: -- Wazuh Indexer --
16/05/2025 16:00:41 INFO: Starting Wazuh indexer installation.
16/05/2025 16:03:44 INFO: Wazuh indexer installation finished.
16/05/2025 16:03:44 INFO: Wazuh indexer post-install configuration finished.
16/05/2025 16:03:44 INFO: Starting service wazuh-indexer.
16/05/2025 16:04:38 INFO: Wazuh indexer service started.
16/05/2025 16:04:38 INFO: Initializing Wazuh Indexer cluster security settings.
16/05/2025 16:04:51 INFO: Wazuh Indexer cluster initialized.
16/05/2025 16:04:51 INFO: -- Wazuh server --
16/05/2025 16:04:51 INFO: Starting the Wazuh manager installation.
16/05/2025 16:07:59 INFO: Wazuh manager installation finished.
16/05/2025 16:07:59 INFO: Starting service wazuh-manager.
16/05/2025 16:08:38 INFO: Wazuh manager service started.
16/05/2025 16:08:38 INFO: Starting Filebeat installation.
16/05/2025 16:08:58 INFO: Filebeat installation finished.
16/05/2025 16:08:58 INFO: Filebeat post-install configuration finished.
16/05/2025 16:08:58 INFO: Starting service filebeat.
16/05/2025 16:08:55 INFO: Filebeat service started.
16/05/2025 16:08:55 INFO: -- Wazuh dashboard --
16/05/2025 16:08:55 INFO: Starting Wazuh dashboard installation.
16/05/2025 16:11:12 INFO: Wazuh dashboard installation finished.
16/05/2025 16:11:12 INFO: Wazuh dashboard post-install configuration finished.
16/05/2025 16:11:12 INFO: Starting service wazuh-dashboard.
16/05/2025 16:11:14 INFO: Wazuh dashboard service started.
16/05/2025 16:12:16 INFO: Initializing Wazuh dashboard web application.
16/05/2025 16:12:18 INFO: Wazuh dashboard web application initialized.
16/05/2025 16:12:18 INFO: -- Summary --
16/05/2025 16:12:18 INFO: You can access the web interface https://wazuh-dashboard-ip:443
User: admin
Password: emq7ao.DQB_pm7GLyY3FWjRR0baktRKE
16/05/2025 16:12:18 INFO: Installation finished.
az12cems@isi:~$ _
```

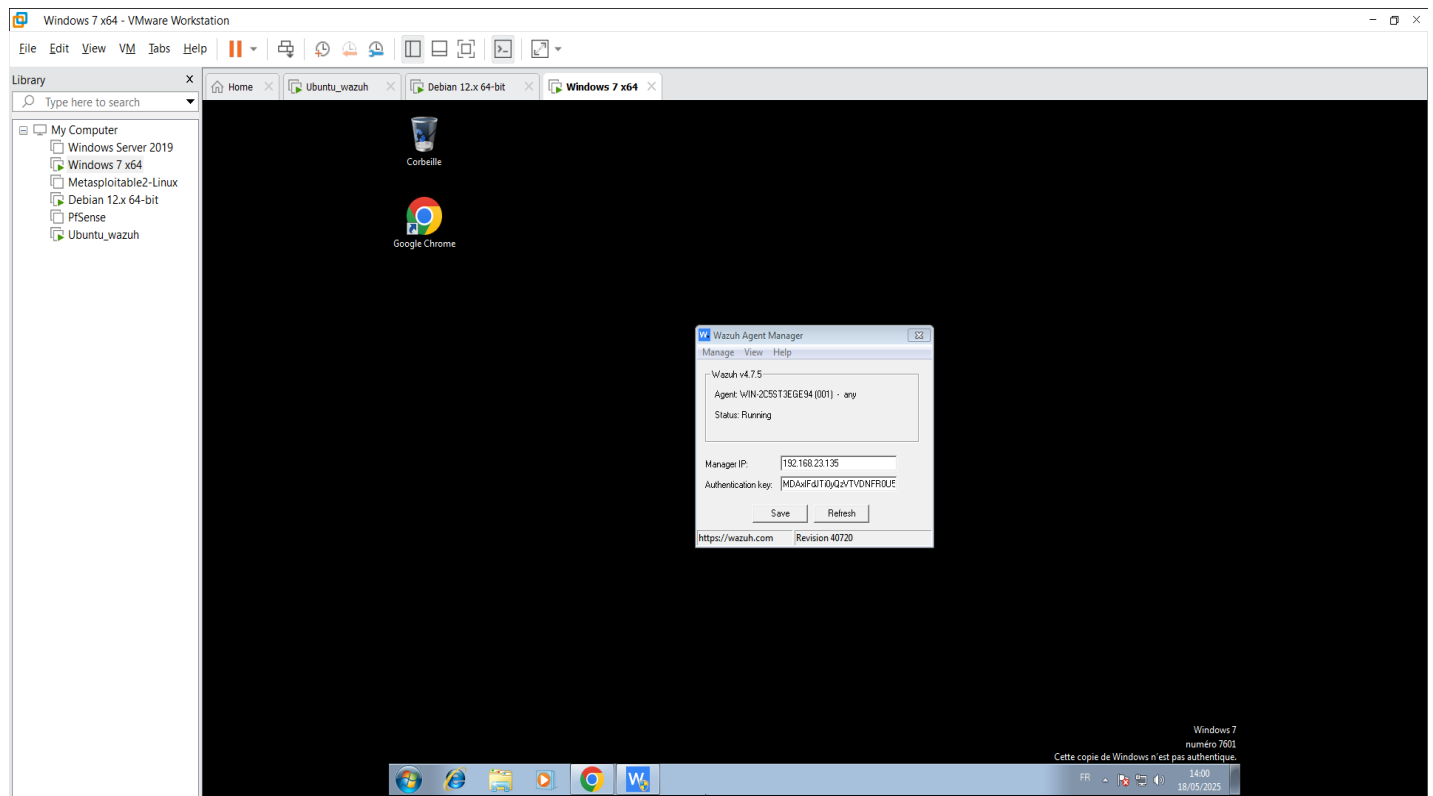
4. Accès au Dashboard Web

- Accès via navigateur : <https://192.168.23.135>
- Port : 443
- Problème rencontré : "Connection was reset"
- Résolution : redémarrage de wazuh-dashboard



5. Installation de l'agent Wazuh sur Windows

- Téléchargement de l'agent Wazuh sur le site officiel
- Configuration :
 - IP du Manager : 192.168.23.135
 - Agent name : générer automatiquement
- Vérification dans le Dashboard : agent "Connected"



6. Génération d'alertes de sécurité

Test 1 : Création de fichier suspect

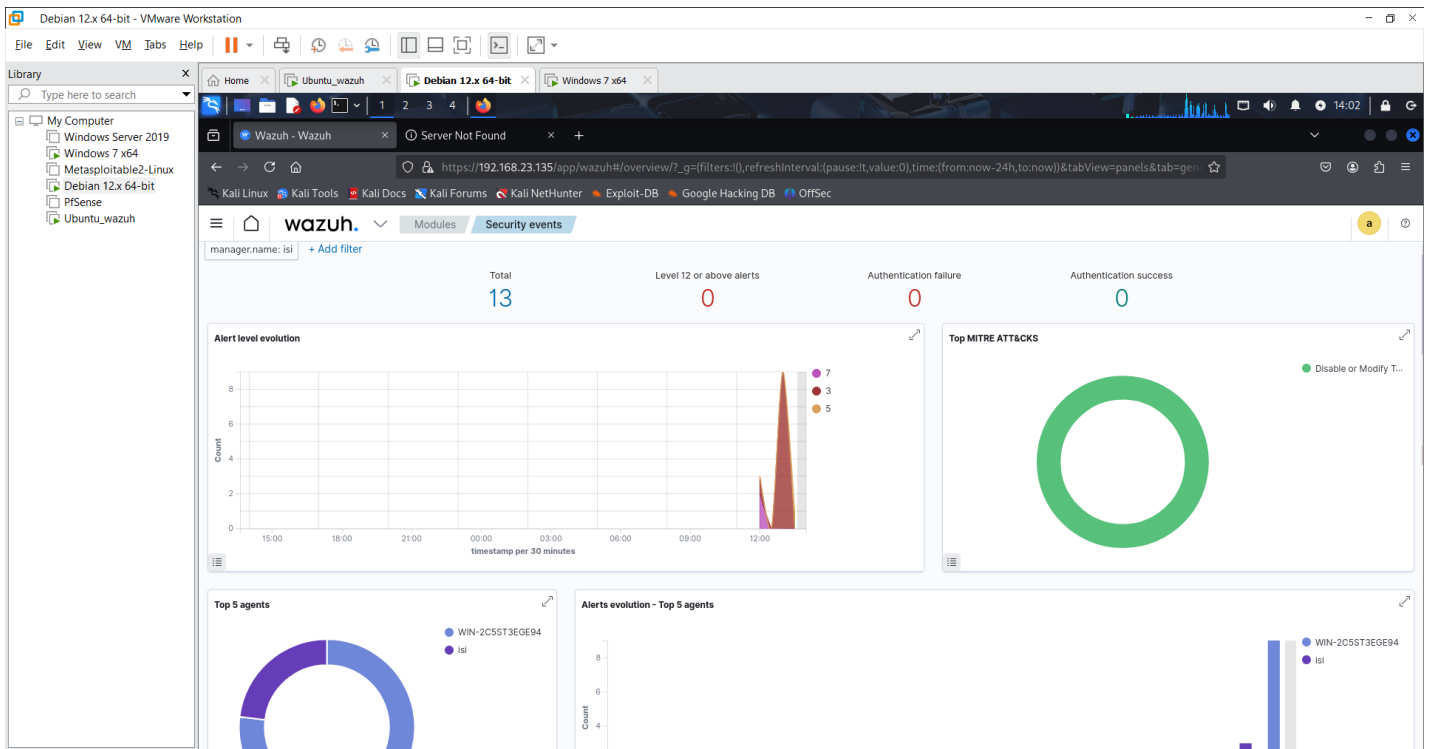
```
echo "malware" > C:\Users\Public\malware.exe
```

Test 2 : Création d'un nouvel utilisateur

```
net user testuser Test@123 /add
```

Test 3 : Tentatives de connexion échouées

→ Résultat : Alertes visibles dans le Dashboard Wazuh



The screenshot shows the Wazuh Security events dashboard with a list of alerts. The table has columns for Time, Agent, Agent name, Technique(s), Tactic(s), Description, Level, and Rule ID. The alerts are sorted by time, showing events from May 18, 2025, at 13:36:00.678 to 13:14:00.746. The alerts include events for name resolution, agent started/stopped, and new agent connection.

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
May 18, 2025 @ 13:36:00.678	001	WIN-2C5ST3EGE94			Name resolution for the name wpad.localdomain timed out	5	61109
May 18, 2025 @ 13:16:08.150	001	WIN-2C5ST3EGE94			Wazuh agent started.	3	503
May 18, 2025 @ 13:16:06.611	001	WIN-2C5ST3EGE94	T1562.001	Defense Evasion	Wazuh agent stopped.	3	506
May 18, 2025 @ 13:15:26.830	001	WIN-2C5ST3EGE94			Wazuh agent started.	3	503
May 18, 2025 @ 13:15:25.131	001	WIN-2C5ST3EGE94	T1562.001	Defense Evasion	Wazuh agent stopped.	3	506
May 18, 2025 @ 13:14:49.158	001	WIN-2C5ST3EGE94			Wazuh agent started.	3	503
May 18, 2025 @ 13:14:47.736	001	WIN-2C5ST3EGE94	T1562.001	Defense Evasion	Wazuh agent stopped.	3	506
May 18, 2025 @ 13:14:13.749	001	WIN-2C5ST3EGE94			Wazuh agent started.	3	503
May 18, 2025 @ 13:14:02.526	001	WIN-2C5ST3EGE94	T1562.001	Defense Evasion	Wazuh agent stopped.	3	506
May 18, 2025 @ 13:14:00.746	001	WIN-2C5ST3EGE94			New wazuh agent connected.	3	501

7. Analyse des logs

- Niveau de sévérité : 5 à 10
- Type d'alertes :
 - Fichier malveillant détecté
 - Événement de gestion des comptes utilisateurs
 - Connexion échouée suspecte.

8. Conclusion

- Le serveur Wazuh a été installé avec succès.
- L'agent Windows est bien connecté et envoie des logs.
- Les alertes sont bien générées et visibles.
- Ce système permet une supervision efficace des postes Windows en environnement local.